FitBuddy

Performance Testing Plan

Proposed test plan based on the supplied FitBuddy project brief

Purpose

Check that FitBuddy remains responsive and reliable while users submit profile details, generate plans, request revisions, and view saved records.

What to measure

Response time: median and 95th percentile for each user action.
Throughput: successful requests per second or minute.
Reliability: error rate, timeouts, and failed AI/database calls.
Resources: CPU, memory, database size, and connection usage.
AI behavior: provider latency, rate limits, and failed/empty responses.

Suggested first-release targets - revise after baseline measurement

Non-AI page loads: p95 at or below 1 second under light test load.
App processing excluding AI wait: p95 at or below 500 ms.
Successful requests excluding provider failures: at least 99% in a short load run.
AI request latency: measure and report separately; it depends on external service.

Test scenarios

Home and form: load pages and submit valid and invalid data.
Workout generation: submit concurrent users and record AI and end-to-end latency.
Feedback update: retrieve a saved plan, request a revision, and save the new version.
Admin view: read a growing user/plan list and monitor database query time.
Failure handling: simulate slow/unavailable AI service and verify bounded waits and clear errors, without losing saved original plans.

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Test Execution and Reporting

Test types

Baseline: one user at a time to establish normal response times.
Load: increase gradually to the expected demo or initial user traffic.
Stress: increase beyond expected load to identify the failure point and recovery.
Soak: maintain modest traffic to detect leaks or slow degradation.

Safe test procedure

1. Use a local/staging environment and synthetic test accounts only.
2. Record app version, machine, database size, AI model, and test settings.
3. Warm up, run each scenario at set concurrency, then repeat for consistency.
4. Separate AI provider wait from app, template, and database processing time.
5. Stop if errors rise sharply or provider limits are approached.

Example load profile

Start with 1 user; ramp to 5, then 10 concurrent users for a short run.
These are example development loads, not a production capacity promise.

Report for each run

Scenario and concurrency; request count; median/p95 latency; throughput; error rate.
AI latency and failures; CPU/memory; database timing; observations and next actions.
Compare results with the proposed targets and previous baseline.

Likely improvement areas

Use database indexes for lookup fields; avoid repeated queries and oversized results.
Set explicit AI timeouts and handle rate limits; keep secrets out of logs.
Do not cache personalized plans across users. Cache only safe, non-personal assets.
Record measurements before optimizing; retest after each material change.

Status

This document is a test plan. No performance tests or results are claimed.